Paper 7: Action

Internalization, the Lagrangian, and the Standard Model
from Finite Admissibility

v2.0

E. S. Brooke

Admissibility Physics — February 2026 (v2.0 full-scope rebuild: April 2026)

Abstract

The Lagrangian of the Standard Model, together with Einstein–Hilbert gravity, a cosmological constant, the Higgs potential, and the one-loop renormalisation-group flow, is derived here from finite admissibility alone. The derivation proceeds through four stages.

First, a structural result from the February 2026 release: zero accumulated action is inadmissible for any path that introduces an irreversible distinction. The minimum ε is the MD (minimum-cost-floor) component of PLEC made quantitative. When accumulated enforcement cost is identified with action in the variational regime of Paper 6, ε is Planck's constant.

Second, the APF partition function $Z_0(\beta) = (1 + d_{\text{eff}}\, e^{-\beta\varepsilon^*})^{C_{\text{total}}}$ is derived elementarily from the binomial identity over admissible subsets of the capacity budget. At $d_{\text{eff}} = 1$ it is Fermi–Dirac; at $d_{\text{eff}} = 102$ it is the Potts-like generalisation forced by $L_{\text{self_exclusion}}$. APF is the ground-state approximation valid for $kT \ll \varepsilon^*$; identified with $\varepsilon^* \sim E_P$, this covers the observable universe by $\sim 10^{32}$ and breaks down at the Planck era. Saturation factorisation gives the thermodynamic formalism Paper 3 v3.0 forward-references for the ledger-accounting principle.

Third, the central structural identity of this paper: the Connes spectral action $S_f = \text{Tr}[f(D/\Lambda)]$ is identical to the APF partition function $Z(\beta) = \text{Tr}[e^{-\beta H}]$ (L_spectral_action_internal). The Seeley–DeWitt expansion of the heat kernel then yields the cosmological constant, Einstein–Hilbert action, gauge kinetic terms, and Higgs potential as separate expansion coefficients. The SM Lagrangian plus gravity is not written down; it is read off the partition function.

Fourth, geometric and group-theoretic internalization: the continuum limit, smooth atlas, Einstein field equations in $d = 4$, direct-product Poincaré $\times\, G_{\text{gauge}}$ structure, and full Lorentzian metric reconstruction from causal order are all derived from APF, eliminating the last external theorem imports of the framework. One-loop beta functions, the Type-I seesaw, spin-statistics, and gauge-group recovery follow as arithmetic and linear-algebra consequences of the APF-derived particle content.

We frame the quantum-of-action theorem as PLEC's MD component made sharp, and the spectral-action identity as PLEC's variational content made explicit. The four structurally necessary PLEC components — A1 (finite upper bound), MD (cost floor), A2 (argmin selection), BW (budget-window non-degeneracy) — each do independent work and are inventoried in Appendix C. Full formal proofs appear in the accompanying Technical Supplement v1.0.

Contents

1 Introduction **4**

1.1 The Problem . . . 4
1.2 The Answer . . . 4
1.3 PLEC framing . . . 4
1.4 What this paper does NOT claim . . . 5

Part I — The Quantum of Action **5**

2 Primitive Objects and Axioms **6**
2.1 Distinctions, Interfaces, Capacity . . . 6
2.2 Enforcement Demand and Cost . . . 6
2.3 Connection to T_ε: the MD component of PLEC . . . 7

3 Admissible Paths and Accumulated Cost **7**

4 Records and Irreversibility **7**
4.1 Definition of Record . . . 7
4.2 Executable Verification: T0.4′ Certificate . . . 7

5 The Minimum Action Theorem **7**
5.1 Lemma: Cost of Introducing a Distinction . . . 7
5.2 Main Theorem: Isolation of Zero . . . 8
5.3 What the Theorem Does NOT Say . . . 8

6 Multiple Records and k! History Sectors **8**
6.1 The n-Record Bound . . . 8
6.2 Connection to k! History Counting (T9) . . . 8

Part II — The APF Partition Function and the Regime of Validity **8**

7 The Non-Interacting APF Partition Function **8**
7.1 Derivation from the binomial identity . . . 8
7.2 The d_{eff} generalisation . . . 9
7.3 Observation on the fermionic-substrate question . . . 9

8 Gapped Ground State: APF's Regime of Validity **9**
8.1 Physical interpretation of β . . . 9
8.2 The observable universe margin . . . 9
8.3 Consistency with $L_{\text{singularity_resolution}}$ . . . 10
8.4 Scope claim, sharpened . . . 10

9 Saturation Factorisation and the Ledger-Accounting Bridge **10**
9.1 The factorisation . . . 10
9.2 Consequence: saturation entropy is a count . . . 10
9.3 The bridge to Paper 3 v3.0's ledger-accounting principle . . . 10
9.4 Status of $\Lambda \cdot G$ . . . 11

Part III — Internalization: the Lagrangian is Derived **11**

10 Spectral Action Equals APF Partition Function **11**
10.1 What this identity means . . . 12
10.2 The A2 component of PLEC in the uniqueness argument . . . 12

11 Scalar-Potential Internalization **12**
11.1 The $\frac{1}{2}$ coefficient from KO-dimension 6 . . . 12
11.2 The scalar potential form from spectral invariance + A1 . . . 12

12 The Enforcement Potential $V(\Phi)$ and the Mass Gap **13**
12.1 Form of the potential . . . 13
12.2 Mass gap and quantum modes . . . 13
12.3 Connection to the spectral-action reading . . . 13

Part IV — Geometric Internalization: Spacetime is Forced **13**

13 Continuum Limit from Finite Capacity and Marginalization **13**

14 Smooth Atlas from Lipschitz Cost **14**

15 Einstein Equations: Uniqueness in $d = 4$ **14**

16 Direct-Product Symmetry Structure **14**

17 Causal Order Determines the Lorentzian Metric **15**

Part V — The Standard Model Lagrangian and Extensions **15**

18 One-Loop Renormalization-Group Flow **15**

19 Neutrino Sector and the Type-I Seesaw **16**

20 Spin-Statistics and Gauge-Group Recovery **16**
20.1 Pauli–Jordan commutator symmetry . . . 16
20.2 McKean–Singer index formula . . . 16
20.3 Gauge group recovery via Tannaka–Krein . . . 16

21 BSM Extensions: Mechanisms from APF Structure **17**
21.1 Dark matter . . . 17
21.2 Inflation . . . 17
21.3 Matter–antimatter asymmetry . . . 17

22 Status of Planck's Constant **17**

23 What Would Falsify This Result **18**

24 Summary **18**
24.1 The central result . . . 18
24.2 Physical implications . . . 19
24.3 Conceptual status . . . 19
24.4 Paper 3 v3.0 bridge . . . 19

A Framework Cost Parameters: ε, κ, η **20**

B Toy-Model Verification of Minimum Action **20**

C PLEC Assumption Inventory for Paper 7 **20**

D Changelog **21**

1. Introduction

1.1 The Problem

Classical mechanics permits arbitrarily small action. Infinitesimal changes in configuration correspond to vanishing accumulated cost. Physical reality does not share this feature. Empirically, action exhibits a minimum scale — Planck's constant $\hbar$ — below which no transitions occur. The usual treatment encodes this minimum axiomatically. Quantum mechanics postulates $\hbar$; it does not explain why a minimum must exist.

Classical mechanics also presents the Lagrangian as something the physicist writes down. The form of the SM Lagrangian — the gauge kinetic terms, the Higgs potential, the Einstein–Hilbert action, the specific Yukawa structure — is something one *chooses*, constrained by symmetry, power-counting renormalisability, and experiment. This is a strong input, even when disguised as a principle. A framework that began with only finite admissibility should, in principle, be able to derive the Lagrangian rather than import it.

1.2 The Answer

This paper shows that zero action is structurally excluded for record-forming transitions under finite admissibility (Part I), and that the SM Lagrangian plus Einstein–Hilbert gravity is the Seeley–DeWitt expansion of the APF partition function (Parts II–III), with the underlying geometry and symmetry group forced by admissibility (Part IV) and the one-loop flow and BSM extensions following arithmetically (Part V).

The argument chain is: (1) Distinctions require enforcement (Paper 1). (2) Enforcement capacity is finite at each interface. (3) Independent distinctions require nonzero enforcement cost. (4) Accumulated cost tracks irreversible commitment. (5) Record-forming transitions cannot have zero cost. (6) In variational regimes, this cost is action. (7) Therefore zero action is inadmissible. The minimum is not postulated; it is derived from finite enforceability.

(8) The APF partition function over admissible capacity subsets factorizes as $Z(\beta) = \mathrm{Tr}[e^{-\beta H}]$ on the finite Hilbert space $(\mathbb{C}^2)^{\otimes 61}$. (9) The Connes spectral action on the APF spectral triple is, by direct computation, identical to this partition function at $\beta = 1/\Lambda^2$. (10) The Seeley–DeWitt small-parameter expansion of the heat kernel produces the cosmological constant, Einstein–Hilbert action, gauge kinetic terms, and Higgs potential as distinct terms with coefficients computed from APF-derived matrices. The Lagrangian is the heat-kernel expansion of the admissibility partition function.

1.3 PLEC framing

This paper sits inside the PLEC rollout. PLEC (the *Principle of Least Enforcement Cost*) is the canonical variational formulation of admissibility, subordinate to the Enforceability of Distinction (Paper 1). It is a statement, not an axiom: admissible evolution is the cost-minimum on a feasible set bounded above by A1 and bounded below by MD. Paper 1 proves that PLEC has four structurally necessary components — any one of which, if removed, produces a degenerate or ill-posed optimization:

- **A1** (*finite upper bound*): enforcement capacity is finite.

- **MD** (*minimum cost floor*): non-trivial distinctions have positive enforcement cost, bounded below by $\varepsilon > 0$.
- **A2** (*argmin selection*): the admissible configuration is the cost-minimum on the feasible set, not a generic element.
- **BW** (*budget-window non-degeneracy*): the feasible set is genuinely two-sided; neither A1 nor MD is saturated everywhere.

Paper 7 establishes two direct payoffs of PLEC's structural content. First, applied to record-forming admissible paths, the MD component forces a strictly positive minimum on accumulated cost — the quantum of action. Second, the A2 component (*argmin* over a cost functional) is what selects the spectral action as the *unique* action satisfying spectral invariance, gauge invariance, and positivity; internalising the Lagrangian is internalising PLEC's variational structure itself.

1.4 What this paper does NOT claim

- *Not that we compute $\hbar$ numerically.* The structural minimum ε is proved to be positive and universal. Its numerical value depends on the specific capacity budget of our universe and is empirical.
- *Not that we derive specific Yukawa values in this paper.* The Yukawa matrix and mass hierarchy live in Paper 4 (`generations.py`). We use the APF-derived particle content as input.
- *Not that we resolve the hierarchy or little-hierarchy problems.* The Higgs potential form is derived; the specific scale $\sigma_0 \sim 246$ GeV comes from $L_{\text{sigma_VEV}}$ and requires the full capacity-budget structure.
- *Not that BSM predictions are quantitative.* Dark matter, inflation, and baryogenesis appear in Part V as mechanisms that follow naturally from APF structure; quantitative predictions require further work and are flagged as such.
- *Not that the spectral-action identity requires noncommutative geometry as a prior commitment.* The identity works in the reverse direction: one computes the APF partition function directly, recognises it as a spectral action by inspection, and inherits the NCG toolkit. NCG notation is used; it is not a logical dependency.

Part I — The Quantum of Action

Operational primitives: where each term is defined and used

Paper 7 takes Paper 1's PLEC/A1 foundation and Paper 2's capacity counting as inputs, then derives the minimum quantum of action, the APF partition function, the identity with Connes' spectral action, and the Standard Model Lagrangian. Five operational primitives carry the argument. This box is a wayfinder, not new content; every claim points to where the term is defined, axiomatized, or derived.

Quantum of action ε^*. The minimum enforcement cost required to form an admissible record — the "smallest unit" of committed capacity at a quantum interface. *Defined:* §2 (primitives). *Derived as floor:* the Minimum Action Theorem (§5).

Isolation from zero: §4. *Related to $\hbar$:* Part V identification table (the numerical value depends on the universe's specific capacity budget and is deliberately not fixed within Paper 7).

Partition function $Z(\beta)$. The APF-level partition function over admissible records: $Z_0(\beta) = (1 + d_{\text{eff}}\, e^{-\beta\varepsilon^*})^{C_{\text{total}}}$ in the non-interacting regime. *Derived in closed form:* §7. *Regime of validity:* gapped-ground-state regime $k_B T \ll \varepsilon^*$, with margin $\sim 10^{32}$ below Planck (§8).

Saturation factorisation. The decomposition $Z_{\text{sat}}(\beta, \eta) = N_{\text{micro}} \cdot \exp(-\beta E_{\text{sat}}(\eta))$, where $N_{\text{micro}} = d_{\text{eff}}^{C_{\text{total}}}$ is a structural count and $E_{\text{sat}}(\eta)$ is a configuration-dependent energy. *Proved:* §9. *Discharges Paper 3's forward-reference* for the thermodynamic formalism of the ledger-accounting principle.

Spectral triple (A, H, D). An algebra A, a Hilbert space H, and a Dirac operator D whose spectrum encodes admissibility cost. *Used under hypothesis H1* (spectral-triple ansatz, flagged in the Technical Supplement red-team). *Load-bearing in the structural pivot:* spectral action equals APF partition function at a Boltzmann cutoff (§10).

Seeley–DeWitt expansion. The asymptotic expansion of the heat-kernel trace of D^2/Λ^2 in inverse powers of Λ. Coefficients $a_0 = 61$, $a_2 \approx 21.985$, $a_4 \approx 87.201$ at the APF parameters. *Derived:* §10. *Produces:* cosmological constant (a_0), Einstein–Hilbert gravity (a_2), Yang–Mills kinetic term and Higgs potential (a_4). This is how the Standard Model Lagrangian is *read off* the APF partition function rather than postulated.

The Technical Supplement makes the comparison to Stinespring–Choi–GKSL, Connes–Marcolli noncommutative geometry, and Lovelock/Coleman–Mandula/HKM/Malament internalization results explicit, and flags three additional hypotheses (H1 spectral-triple ansatz, H2 Lipschitz cost regularity, H3 HKM future-distinguishing causal order) at point of use.

2. Primitive Objects and Axioms

2.1 Distinctions, Interfaces, Capacity

Definition 2.1 (Distinction)**.** A distinction d is an element of a set D representing a claim that two alternatives are non-identical.

Definition 2.2 (Interface)**.** An interface Γ is a locus at which enforcement is evaluated.

Definition 2.3 (Finite capacity)**.** Each interface i carries finite capacity: $0 < C_i < \infty$ (Axiom A1).

2.2 Enforcement Demand and Cost

Each distinction d imposes demand $\delta_i(d) \geq 0$ at interface i. A set S of distinctions is admissible at interface i if $\sum_{d \in S} \delta_i(d) \leq C_i$. The enforcement cost vector $\delta(d) = (\delta_1(d), \ldots, \delta_n(d))$ captures the multi-interface demand profile.

2.3 Connection to T_ε: the MD component of PLEC

The abstract condition of uniform finite distinguishability — $\|\delta(d)\|_1 \geq \varepsilon > 0$ for every admissible d — is formally identified with the *minimum cost floor* (MD) component of PLEC, and quantitatively with the proved theorem T_ε of Paper 3 (§3.3). T_ε derives from A1 together with meaning-equals-robustness that $\mathrm{cost}(O) = 0$ or $\mathrm{cost}(O) \geq \varepsilon$. If infinitesimal commitments were permitted, finite capacity C could support unbounded distinctions, destroying A1's physical content. The abstract ε of this paper and the proved ε of T_ε are the same quantity; both are the MD component of PLEC. *Code reference:* `check_T_eps` in `core.py`.

3. Admissible Paths and Accumulated Cost

An admissible path is a parameterised family $P = \{S_\lambda\}_{\lambda\in[0,1]}$ of admissible distinction sets. The accumulated cost (action) of P is defined via total variation:

$$A(P) \;=\; \sum_i \mathrm{Var}_\lambda\Big[\textstyle\sum_{d\in S_\lambda} \delta_i(d)\Big]\,.$$

This tracks how much enforcement load changes along the path, summed over all interfaces.

Figure 1 is the visual that the rest of Part I makes formal: a record (§4) is a persistent point in the locked subspace; the Minimum Action Theorem (§5) puts a positive floor on the cost of moving through the narrowing; the $k!$ history sectors (§6) are the three tubes after locking. The diagram is the static picture; this paper supplies its dynamics.

4. Records and Irreversibility

4.1 Definition of Record

Definition 4.1 (Record)**.** A distinction d is a record at admissible set S if (i) $d \in S$ (currently enforced) and (ii) there is no admissible path from S to $S \setminus \{d\}$ (cannot be removed without violating admissibility). Records are persistent enforcement commitments.

4.2 Executable Verification: T0.4′ Certificate

A machine-checkable proof that records exist is provided by the T0.4′ BFS certificate of Paper 2 (Appendix). It operates on the canonical witness world $W = (\{a, b, c, h, r\}, \Gamma_1, \Gamma_2)$. Removing record r requires handoff token h, but $E(r) + E(h) = 2 + 9 = 11 > 10 = C_{\Gamma_1}$. Every attempted removal passes through an inadmissible state. The BFS exhaustively verifies: no admissible path from S to $S \setminus \{r\}$ exists. Record r is operationally irreversible — by capacity constraint, not by fiat. *Code reference:* `check_T0_4_prime_BFS` in `paper1.py`.

5. The Minimum Action Theorem

5.1 Lemma: Cost of Introducing a Distinction

Lemma 5.1. *Let $P = \{S_\lambda\}$ be an admissible path. Suppose d is absent from S_{λ^-} and present in S_{λ^+}, and d is independent of S_{λ^-}. Then $A(P) \geq \|\delta(d)\|_1 \geq \varepsilon$.*

Proof sketch. For each interface i, the function $f_{i,d}(\lambda) = \mathbf{1}_{d\in S_\lambda} \cdot \delta_i(d)$ changes from 0 to $\delta_i(d)$ over $[\lambda^-, \lambda^+]$, so $\mathrm{Var}_\lambda[f_{i,d}] \geq \delta_i(d)$. Summing: $A(P) \geq \sum_i \delta_i(d) = \|\delta(d)\|_1 \geq \varepsilon$ by T_ε. Full proof

in Technical Supplement §S.1. □

5.2 Main Theorem: Isolation of Zero

Theorem 5.2 (Isolation of Zero Action, **[P]**). *Assume A1 (finite admissibility) and MD (uniform finite distinguishability, $= T_\varepsilon$). Let $\mathcal{S} = \{A(P) : P$ is a record-forming admissible path$\}$. Then $\inf \mathcal{S} \geq \varepsilon > 0$; zero is isolated from $\mathcal{S}$.*

Proof sketch. Any record-forming path introduces at least one independent distinction d that becomes a record. By Lemma 5.1, $A(P) \geq \varepsilon > 0$. Therefore every element of $\mathcal{S}$ is at least ε. Zero is isolated. □

5.3 What the Theorem Does NOT Say

1. Not all paths have minimum cost — reversible reallocations may have $A = 0$.
2. Action is not discrete — values above ε form a continuum.
3. The bound is not necessarily tight — $\inf \mathcal{S}$ may exceed ε.
4. The numerical value is not determined — ε depends on physical realization.

6. Multiple Records and k! History Sectors

6.1 The n-Record Bound

Proposition 6.1. *If a path introduces n independent records, then $A(P) \geq n \cdot \varepsilon$.*

Each independent record contributes at least ε by Lemma 5.1; the contributions sum by path additivity.

6.2 Connection to k! History Counting (T9)

Paper 5, §7.3, establishes that k record-locking enforcement operations applied in all $k!$ orderings yield $k!$ orthogonal record sectors. Combining with Proposition 6.1: the total action cost of creating $k!$ distinguishable histories is $A \geq k \cdot \varepsilon$. The entropy contribution is $\Delta S_{\text{records}} \geq \log(k!) \sim k \log k$. This connects the action minimum to the entropy accounting of Paper 3: each ε of action invested creates one record, and the factorial growth of distinguishable histories ($k!$) comes from the combinatorics of irreversible ordering. *Code reference:* `check_T9` in `core.py`.

Part II — The APF Partition Function and the Regime of Validity

7. The Non-Interacting APF Partition Function

7.1 Derivation from the binomial identity

APF's feasible configurations at an interface are subsets of $C_{\text{total}} = 61$ candidate capacity types, each contributing cost ε^* if included. At the non-interacting level (ignoring the interference surplus Δ from L_{nc}), the Gibbs partition function over the admissible set is:

$$Z_0(\beta) = \sum_{\text{admissible}} e^{-\beta \cdot \text{cost}} = \sum_{k=0}^{C_{\text{total}}} \binom{C_{\text{total}}}{k} e^{-\beta k \varepsilon^*} = \left(1 + e^{-\beta \varepsilon^*}\right)^{C_{\text{total}}}.$$

The last equality is the binomial identity. This is an elementary calculation. What is notable is its shape: the APF partition function is exactly the partition function of C_{total} non-interacting fermionic modes with single-particle energy ε^*.

7.2 The d_{eff} generalisation

$L_{\text{self_exclusion}}$ forces $d_{\text{eff}} = 102$ internal states per mode. The partition function generalises to

$$Z_0(\beta) \;=\; \left(1 + d_{\text{eff}}\, e^{-\beta\varepsilon^*}\right)^{C_{\text{total}}},$$

which is the Potts-like multi-level occupation analogue. At $d_{\text{eff}} = 1$ this reduces to Fermi–Dirac form; at $d_{\text{eff}} = 102$ it supports the saturation entropy $S_{\text{dS}} = C_{\text{total}} \log d_{\text{eff}} = 61 \log 102 \approx 282.12$ nats. *Code reference:* `check_L_self_exclusion` in `gravity.py`.

7.3 Observation on the fermionic-substrate question

The non-interacting partition function has the same mathematical form as fermionic occupation. The 45 Weyl fermions of the SM matter content sit naturally on the 19 "matter" modes of the $42 + 19 = 61$ partition, and the 42 "vacuum" modes correspond to unoccupied states. The structural coincidence is suggestive — fermionic SM matter content is consistent with an underlying binary-distinction substrate — but we are careful not to claim a new derivation from the partition-function reading alone. T_{field} derives the 45 Weyl fermions through the independent gauge/Lorentz representation-theory chain; the partition-function observation is a structural consistency check, not a replacement derivation. We keep this as an observation.

8. Gapped Ground State: APF's Regime of Validity

8.1 Physical interpretation of β

β in the APF partition function has a clean physical interpretation: inverse thermal temperature in units of $1/\varepsilon^*$. The APF operational regime is $\beta\varepsilon^* \gg 1$, equivalently

$$kT \;\ll\; \varepsilon^*.$$

Identifying ε^* with a Planck-scale energy (the natural scale for gravitational physics; see L_{cost}), the condition $kT \ll \varepsilon^*$ translates numerically as follows.

8.2 The observable universe margin

For the observable universe at $kT \sim 2.7$ K $\sim 10^{-4}$ eV and $\varepsilon^* \sim E_P \sim 10^{28}$ eV:

$$\beta\varepsilon^* \;\sim\; 10^{32}.$$

APF's ground-state approximation is valid with enormous margin. APF breaks down precisely when $kT \sim \varepsilon^*$, which at Planck scale means $T \sim 10^{32}$ K, corresponding to the first Planck time after the Big Bang. This is exactly the regime where quantum gravity is expected to become essential.

8.3 Consistency with $L_{\text{singularity_resolution}}$

The framework's own $L_{\text{singularity_resolution}}$ theorem is consistent with this scope: APF begins at $k = 1$ (minimum de Sitter patch), not at $k = 0$, because the singularity would be in APF's breakdown regime. The partition-function picture and the singularity-resolution picture are two views of the same APF–QG boundary. *Code reference:* `check_L_singularity_resolution` in `cosmology.py`.

8.4 Scope claim, sharpened

APF is the ground-state approximation that holds throughout the observable universe and fails only at the Planck era. This is not an aspirational universalism but a precise statement of where the framework applies. It is also a falsifiable prediction: any observable phenomenon where APF fails outside this window would falsify the ground-state-approximation scope claim.

9. Saturation Factorisation and the Ledger-Accounting Bridge

9.1 The factorisation

At saturation ($k = C_{\text{total}} = 61$, the structurally realised APF configuration), the partition function restricts to the single configuration with all modes occupied. Introducing the interaction corrections $\eta(i,j)$ from L_{nc} superadditivity, the saturation partition function factorizes:

$$Z_{\text{sat}}(\beta,\eta) \;=\; N_{\text{micro}} \cdot \exp\!\big(-\beta E_{\text{sat}}(\eta)\big),$$

with

$$N_{\text{micro}} = d_{\text{eff}}^{C_{\text{total}}}, \qquad E_{\text{sat}}(\eta) = C_{\text{total}}\varepsilon^* + \sum_{i<j} \eta(i,j).$$

N_{micro} is a pure count — β- and η-independent. $E_{\text{sat}}(\eta)$ is an energy shift that absorbs the interaction corrections additively.

9.2 Consequence: saturation entropy is a count

The saturation entropy

$$S_{\text{dS}} \;=\; \log N_{\text{micro}} \;=\; C_{\text{total}} \log d_{\text{eff}} \;=\; 61 \log 102 \;\approx\; 282.12 \text{ nats}$$

inherits independence from both β and the interaction corrections η. Counts do not depend on energy scales, and conditioning on a single mode configuration leaves only the internal d_{eff} multiplicity as the variable degree of freedom. We state this consequence as a single sentence rather than as a named theorem, because it is an immediate algebraic consequence of the factorisation rather than an independent mathematical result. *Code reference:* `check_T_deSitter_entropy` in `gravity.py`.

9.3 The bridge to Paper 3 v3.0's ledger-accounting principle

Paper 3 v3.0, §5 (ledger-accounting principle) establishes that entropy is a count of admissible configurations and that this count is robust under cost-landscape corrections. It forward-references Paper 7 for the thermodynamic formalism. The Z_{sat} factorisation is that formalism. The robustness of entropy under thermal and interaction corrections is the factorisation made ex-

plicit: corrections shift energies without changing multiplicities. The bridge is cleanly two-way: Paper 3 states the principle qualitatively; this paper backs it quantitatively.

9.4 Status of $\Lambda \cdot G$

The holographic identity gives

$$\Lambda \cdot G \;=\; 3\pi \cdot e^{-S_{\mathrm{dS}}} \;=\; 3\pi/102^{61}.$$

Each factor in the count — $C_{\mathrm{total}} = 61$, $d_{\mathrm{eff}} = 102$ — is itself structurally forced (L_{count}, $L_{\mathrm{self_exclusion}}$). $\Lambda \cdot G$ is therefore forced by counting, not tuned by any free parameter. We resist the framing "exact, not approximate," which invites a corrections-dissection attack; the accurate framing is that $\Lambda \cdot G$ is an elementary count of structurally-forced integers.

Part III — Internalization: the Lagrangian is Derived, Not Written Down

10. Spectral Action Equals APF Partition Function

This is the structural pivot of the paper.

Theorem 10.1 (Spectral action = APF partition function, $L_{\mathrm{spectral_action_internal}}$, **[P]**). *The Connes spectral action $S_f = \mathrm{Tr}[f(D_F/\Lambda)]$ on the APF spectral triple is identical to the APF partition function $Z(\beta) = \mathrm{Tr}[e^{-\beta H}]$ from $L_{\mathrm{quantum_evolution}}$, evaluated at $\beta = 1/\Lambda^2$ with the Boltzmann cutoff $f(x) = e^{-x}$.*

Four-step derivation. **Step 1 (partition function = heat kernel).** $Z(\beta) = \mathrm{Tr}[e^{-\beta H}]$ on $\mathcal{H} = (\mathbb{C}^2)^{\otimes 61}$ with the capacity Hamiltonian $H = -\varepsilon^* \sum_i n_i$. The finite Dirac operator D_F from $L_{\mathrm{ST_Dirac}}$ satisfies $D_F^2 = M_Y^\dagger M_Y$ (Yukawa matrix squared). The heat kernel $K(t) = \mathrm{Tr}[e^{-tD_F^2}] = \sum_i e^{-t\lambda_i^2}$ is a finite sum over eigenvalues (no UV divergence).

Step 2 (spectral action = Laplace transform of heat kernel). For any admissible cutoff function f,

$$S_f \;=\; \mathrm{Tr}\big[f(D_F^2/\Lambda^2)\big] \;=\; \int_0^\infty \tilde{f}(t/\Lambda^2)\, K(t)\, \frac{dt}{t},$$

where $\tilde{f}$ is the Laplace–Mellin transform of f. For $f(x) = e^{-x}$, $S_f = K(1/\Lambda^2) = Z(1/\Lambda^2)$. The spectral action is exactly the APF partition function at $\beta = 1/\Lambda^2$.

Step 3 (heat-kernel expansion = physics). The small-t (high-Λ) asymptotic expansion is

$$K(t) \;=\; a_0 - t \cdot a_2 + \frac{t^2}{2} \cdot a_4 + O(t^3),$$

with APF-computed coefficients

$$a_0 = \dim(\mathcal{H}_F) = 96, \qquad a_2 = \mathrm{Tr}(D_F^2) = 21.985, \qquad a_4 = \mathrm{Tr}(D_F^4) = 87.201.$$

In curved spacetime, these multiply the geometric Seeley–DeWitt coefficients to give:

- $f_4 \Lambda^4\, a_0 \;\to$ cosmological constant.
- $f_2 \Lambda^2\, a_2 \;\to$ Einstein–Hilbert action (gravity).

- $f_0\, a_4 \;\rightarrow$ gauge kinetic terms + Higgs potential.

Step 4 (uniqueness). The spectral action is the unique action satisfying: (a) spectral invariance — depends only on eigenvalues of D (the APF cost metric, L_{loc}); (b) gauge invariance — invariant under $A \to UAU^\dagger$ (the APF gauge group, T_{gauge}); (c) positivity — $S > 0$ for $f > 0$ (the APF Boltzmann weight, positivity of $e^{-\beta H}$). Full proof in Technical Supplement §S.3. □

Code reference: `check_L_spectral_action_internal` in `internalization.py`.

10.1 What this identity means

Once the APF partition function is accepted, the SM Lagrangian plus gravity is not written down — it is read off. The Lagrangian is the heat-kernel expansion of the admissibility partition function. Six historical external inputs — Connes–Lott (1991), Chamseddine–Connes–Marcolli (2007), Chamseddine–Connes (2012), and three related references — which were previously logical dependencies of the Lagrangian derivation in prior spectral-action approaches, become *attributions* rather than *imports*. They credit the notation and framework; the theorem content is APF-internal.

10.2 The A2 component of PLEC in the uniqueness argument

Step 4 of the derivation is where PLEC's A2 component (argmin selection) does its work. The spectral action is not one action among many; it is the *unique* action satisfying the three structural conditions. PLEC selects it. Internalising the Lagrangian is internalising the variational structure of PLEC itself.

11. Scalar-Potential Internalization

11.1 The $\frac{1}{2}$ coefficient from KO-dimension 6

Previous spectral-action approaches had to import the factor of $\frac{1}{2}$ on $\mathrm{Tr}(\kappa^\dagger\kappa)$ in the fermion-bilinear part of the action from Chamseddine–Connes. $L_{\text{normalization_coefficient}}$ derives this coefficient.

Lemma 11.1 ($L_{\text{normalization_coefficient}}$, **[P]**). *The factor of $\frac{1}{2}$ on $\mathrm{Tr}(\kappa^\dagger\kappa)$ follows from the KO-dimension 6 of the APF spectral triple. The real structure J satisfies $J^2 = -I$ (from $L_{\text{ST_Dirac}}$ and T_{CPT}), which halves the naive trace because of the antisymmetry between a state and its J-partner.*

Code reference: `check_L_normalization_coefficient` in `internalization.py`.

Full proof in Technical Supplement §S.3.2. The point is that the $\frac{1}{2}$ is not a convention; it is forced by the APF real structure (CPT).

11.2 The scalar potential form from spectral invariance + A1

Lemma 11.2 ($L_{\text{scalar_potential_form}}$, **[P]**). *The specific quartic form of $V(H, \sigma)$ in the scalar sector — the combination of mass terms, quartic self-coupling, and Higgs–singlet coupling that appears in the Higgs potential — is derived from (i) spectral invariance (the action depends only on eigenvalues of D_F) plus (ii) A1 (finite capacity bounds the polynomial degree; higher-than-quartic terms require unbounded energies and violate A1).*

Code reference: `check_L_scalar_potential_form` in `internalization.py`.

The Higgs potential form is not postulated. A spectrally-invariant, A1-bounded polynomial in the scalar sector can only be the specific quartic form that appears in the SM. Together with $L_{\text{normalization_coefficient}}$, these two results upgrade $L_{\text{sigma_VEV}}$ from **[P_structural]** to **[P]**: the Higgs VEV is derived, not merely structurally necessary.

12. The Enforcement Potential $V(\Phi)$ and the Mass Gap

12.1 Form of the potential

Paper 3, §5.3, derives the enforcement cost functional

$$V(\Phi) \;=\; \varepsilon\Phi - \frac{\eta}{2\varepsilon}\Phi^2 + \frac{\varepsilon\Phi^2}{2(C-\Phi)}.$$

V has a binding well at $\Phi/C \approx 0.81$ with mass gap $d^2V/d\Phi^2 = 7.33 > 0$. *Code reference:* `check_L_V_enforcement` in `core.py`.

12.2 Mass gap and quantum modes

The mass gap $d^2V/d\Phi^2 > 0$ implies stable excitations around the binding well. Paper 5 shows no classical solitons localise in the APF Hilbert-space structure, so the excitations are quantum modes. The minimum action ε from Theorem 5.2 sets the scale at which these modes become meaningful: excitations with $S \sim \varepsilon$ exhibit quantum interference; $S \gg \varepsilon$ is classical.

12.3 Connection to the spectral-action reading

Under the spectral-action identity (Theorem 10.1), $V(\Phi)$ is part of the Seeley–DeWitt a_4 coefficient. This connects the microscopic enforcement-potential story of Paper 3 with the macroscopic Lagrangian story of this paper. The enforcement potential is the field-theory realisation of the action bound — not a separate mechanism but the same structure at different scales.

Part IV — Geometric Internalization: Spacetime is Forced

13. Continuum Limit from Finite Capacity and Marginalization

Lemma 13.1 ($L_{\text{kolmogorov_internal}}$, **[P]**). *The continuum limit of the enforcement lattice exists and is unique, derived entirely from A1 (finite capacity) and the marginalisation consistency condition R3.*

Code reference: `check_L_kolmogorov_internal` in `internalization_geo.py`.

Sketch

A1 bounds enforcement capacity finitely in any region; any finite region contains finitely many enforcement events. At resolution ε, the lattice has $N(\varepsilon) \sim (L/\varepsilon)^d$ sites with $d = 4$ (from T_8). Δ_{ordering} derives R3: for any two resolutions $\varepsilon_1 > \varepsilon_2$, the coarse distribution is the marginal of the fine distribution. This is the Kolmogorov consistency condition; the Kolmogorov extension theorem (1933) then guarantees a unique continuum-limit measure. $L_{\text{kolmogorov_internal}}$ proves

the extension's content without citing Kolmogorov — the continuum measure is constructed internally by exhausting subsets at each resolution. Full proof in Technical Supplement §S.4.1.

14. Smooth Atlas from Lipschitz Cost

Lemma 14.1 ($L_{\text{chartability}}$, **[P]**). *The continuum enforcement space admits a smooth (C^∞) atlas, making it a smooth manifold.*

Code reference: `check_L_chartability` in `internalization_geo.py`.

Sketch

The enforcement cost is a metric (identity, symmetry from time reversal, triangle inequality from L_{cost}, non-degeneracy from A1). A1 bounds diameter at $C_{\text{total}} \cdot \varepsilon^*$. The Finite Boundary Condition Δ_{fbc} gives a uniform Lipschitz bound on the enforcement field; combined with elliptic regularity of the Euler–Lagrange equation for the enforcement action, Schauder estimates bootstrap $C^{0,1} \to C^{1,\alpha} \to C^\infty$. Nash–Kuiper and Palais manifold theorems are not imported; the atlas is constructed internally from the APF metric and the enforcement PDE. Full proof in Technical Supplement §S.4.2.

15. Einstein Equations: Uniqueness in $d = 4$

Lemma 15.1 ($L_{\text{lovelock_internal}}$, **[P]**). *In $d = 4$, the gravitational field equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ is the unique response law satisfying the admissibility conditions A9.1–A9.5.*

Code reference: `check_L_lovelock_internal` in `internalization_geo.py`.

Sketch

The five conditions A9.1–A9.5 are each APF-derived: (A9.1) locality from L_{loc}; (A9.2) symmetry from the polarisation identity (T_{7B}); (A9.3) divergence-free from Noether applied to diffeomorphism invariance (contracted Bianchi is a mathematical identity); (A9.4) second-order from A1 (finite capacity $\Rightarrow$ bounded cost $\Rightarrow$ field equation at most second-order in metric derivatives); (A9.5) correct Newtonian limit from $\nabla^2\Phi = 4\pi G\rho$.

Any symmetric, divergence-free, second-order tensor built from $g_{\mu\nu}$ and its first two derivatives in $d = 4$ is a linear combination of $g_{\mu\nu}$, $R_{\mu\nu}$, and $Rg_{\mu\nu}$ (by index counting). The combination is forced: $G_{\mu\nu} + \Lambda g_{\mu\nu}$. Lovelock's 1971 theorem is reproduced; the proof is APF-internal. Full proof in Technical Supplement §S.4.3.

16. Direct-Product Symmetry Structure

Lemma 16.1 ($L_{\text{coleman_mandula_internal}}$, **[P]**). *The full symmetry group of the framework factorizes as $G = \text{Poincaré} \times G_{\text{gauge}}$.*

Code reference: `check_L_coleman_mandula_internal` in `internalization_geo.py`.

Sketch

Two independent derivation chains share only A1: *Chain A* (spacetime: A1 $\to L_{\text{irr}} \to \Delta_{\text{ordering}} \to$ continuum $\to$ signature $\to T_8 \to T_{9,\text{grav}}$) uses irreversibility and causal order; *Chain B* (gauge: A1

$\to L_{\text{loc}} \to T_3 \to T_{\text{gauge}}$) uses locality and superselection. Locality requires $[G_S(x), G_I(y)] = 0$ for spacelike-separated points. Combined, this forces G to be a direct product. Coleman–Mandula (1967) and Haag–Łopuszański–Sohnius (1975) are reproduced; the proof is APF-internal. Full proof in Technical Supplement §S.4.4.

17. Causal Order Determines the Lorentzian Metric

Lemma 17.1 ($L_{\text{HKM_causal_geometry}}$, **[P]**). *The partial order $\preceq$ on events from L_{irr} uniquely determines the conformal class of the Lorentzian metric $[g_{\mu\nu}]$.*

Lemma 17.2 ($L_{\text{Malament_uniqueness}}$, **[P]**). *Causal order plus A1's volume normalization uniquely determines the full metric $g_{\mu\nu}$.*

Code reference: `check_L_HKM_causal_geometry, check_L_Malament_uniqueness` in `extensions.py`.

Sketch

L_{irr} gives a strict partial order on events satisfying transitivity, irreflexivity, and denseness. Causal cones at each point coincide with null cones; the tangent structure forced by mapping null vectors to null vectors is conformal. HKM (1976) then gives $[g_{\mu\nu}]$. Malament (1977) shows a causal isomorphism is a conformal isometry: $f^* g_2 = \Omega^2 g_1$. A1 provides a volume measure (the capacity budget normalised over the continuum manifold), which fixes Ω^2. The full metric is determined. Paper 6 will provide a dynamical treatment; the self-contained version here makes Paper 7's Lagrangian derivation independent of forward references. Full proofs in Technical Supplement §S.4.5.

Part V — The Standard Model Lagrangian and Extensions

18. One-Loop Renormalization-Group Flow

Theorem 18.1 ($T_{6B,\text{beta_one_loop}}$, **[P]**). *The one-loop beta coefficients for the SM gauge couplings, top Yukawa, and Higgs quartic self-coupling are derived by Casimir/Dynkin arithmetic applied to the APF-derived particle content T_{field} and T_{gauge}, with $N_{\text{gen}} = 3$ from T_7.*

Code reference: `check_T6B_beta_one_loop` in `extensions.py`.

Content

The one-loop formula $\beta(g_i) = -b_i g_i^3/(16\pi^2)$ has b_i depending only on gauge-group Casimirs, Dynkin indices of matter representations, and N_{gen}. Plugging the APF-derived SM content per generation — Q_L, u_R, d_R, L_L, e_R, H — into the general formula

$$b_{SU(N)} \;=\; (11/3)C_2(\text{adj}) - (2/3)\textstyle\sum_f T(R_f) - (1/3)\sum_s T(R_s)$$

yields the standard SM values $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$. The beta functions are not imported; they are arithmetic consequences of the APF-derived spectrum. Full derivation in Technical Supplement §S.5.1.

19. Neutrino Sector and the Type-I Seesaw

Lemma 19.1 ($L_{\text{seesaw_type_I}}$, **[P]**). *The APF-derived Dirac operator D_F with the Majorana block (from $L_{\text{nuR_enforcement}}$) yields the type-I seesaw mass formula $m_\nu = -M_D M_R^{-1} M_D^T$.*

Code reference: `check_L_seesaw_type_I` in `extensions.py`.

Content

The 6×6 neutral-fermion mass matrix in the (ν_L, ν_R) basis is

$$M_{\text{neutral}} = \begin{pmatrix} 0 & M_D \\ M_D^T & M_R \end{pmatrix},$$

with M_D (Dirac) 3×3 and M_R (Majorana) 3×3 symmetric. For $M_R \gg M_D$ (hierarchy from $L_{\text{dm2_hierarchy}}$), perturbative block diagonalization yields $m_{\text{light}} = -M_D M_R^{-1} M_D^T$ and $m_{\text{heavy}} \approx M_R$. Light neutrino masses scale as m_D^2/M_R. Full five-step derivation in Technical Supplement §S.5.2.

20. Spin-Statistics and Gauge-Group Recovery

20.1 Pauli–Jordan commutator symmetry

Lemma 20.1 ($L_{\text{Pauli_Jordan}}$, **[P]**). *The Pauli–Jordan commutator function satisfies the reflection symmetry required for causal propagation and spin-statistics. Derived internally.*

Code reference: `check_L_Pauli_Jordan` in `extensions.py`.

20.2 McKean–Singer index formula

Lemma 20.2 ($L_{\text{McKean_Singer_internal}}$, **[P]**). *The McKean–Singer supertrace formula* $\text{Index}(D) = \text{Tr}_s[e^{-tD^2}] = \text{Tr}[e^{-tM^\dagger M}] - \text{Tr}[e^{-tMM^\dagger}]$ *is proved from pure linear algebra (SVD), eliminating the last external theorem import in the anomaly-cancellation chain.*

Code reference: `check_L_McKean_Singer_internal` in `extensions.py`.

Full five-step proof (SVD decomposition, eigenvalue identity, supertrace cancellation, limit $t \to 0$, matching to APF index) in Technical Supplement §S.5.3.

20.3 Gauge group recovery via Tannaka–Krein

Lemma 20.3 ($L_{\text{Tannaka_Krein}}$, **[P]**). *The APF-derived symmetric C^*-tensor category of localised superselection sectors satisfies the four Tannaka–Krein conditions. The compact group $G = \text{Aut}(\omega)$ recovered from the fiber functor ω is the SM gauge group* $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$.

Code reference: `check_L_Tannaka_Krein` in `extensions.py`.

Four Tannaka–Krein conditions from APF

(TK1) *Monoidal:* L_{loc} provides a net of local algebras; composition of localised superselection sectors gives $\otimes$.

(TK2) *Symmetric $\epsilon^2 = 1$:* $T_{\text{spin_statistics}}$ with $d_{\text{space}} = 3$ forces $\pi_1(\text{Config}) = S_n$, so statistics phases satisfy $\kappa^2 = 1$ (no anyons in $d = 4$).

(TK3) *Conjugates:* CPT provides dual objects with unit and counit.

(TK4) *Fiber functor:* the vacuum sector provides $\omega : \mathcal{C} \to \mathrm{Vect}_{\mathbb{C}}$, monoidal by locality.

Doplicher–Roberts (1989) is reproduced; the proof is APF-internal. This replaces the DR import in T_3. Full proof in Technical Supplement §S.5.4.

21. BSM Extensions: Mechanisms from APF Structure

Three BSM directions follow naturally from the internalized framework. We name the mechanisms and relevant code references; we do not claim specific quantitative predictions.

21.1 Dark matter

The APF enforcement budget admits lightweight sterile states as cost-minimizers in unoccupied regions of capacity space. T_{12} in Paper 5 derives the existence and basic properties of such states. Particle-level identification remains open; no specific mass or cross-section is predicted here. The mechanism is: the capacity budget is not filled by SM matter alone; residual unoccupied capacity supports states that couple weakly (by L_{loc} restricted to the unoccupied sector) and evade direct detection.

21.2 Inflation

The pre-saturation regime ($k < C_{\mathrm{total}} = 61$) has the staged-relaxation structure developed in Paper 3 v3.0, §(Cosmogenesis). The enforcement-cost profile during staged constraint relaxation provides a natural early-time epoch compatible with inflationary phenomenology: constraint relaxation stages release capacity on scale-dependent timescales, producing a nearly-scale-invariant power spectrum with specific stage-dependent deviations. Full dynamical treatment and comparison with CMB observations await Paper 6 and dedicated inflation-staircase work.

21.3 Matter–antimatter asymmetry

The irreversibility structure L_{irr} gives rise to two distinct CP-violation mechanisms: *holonomy phases* in the CKM sector (derived in `generations.py`) and *entropy-optimization phases* in the PMNS sector (derived in `session_delta_pmns.py`). Both mechanisms derive from the same irreversibility principle. Baryogenesis-level quantitative predictions require further work connecting these CP phases to the Sakharov conditions via sphaleron/electroweak dynamics.

22. Status of Planck's Constant

Planck's constant $\hbar$ is the empirical value of the structural minimum ε:

$$\hbar \;=\; (\text{physical instantiation of } \varepsilon) \;\approx\; 1.055 \times 10^{-34}\ \mathrm{J} \cdot \mathrm{s}.$$

The framework explains why $\varepsilon > 0$ must exist (structural necessity from MD + A1) and why it is universal (applies to all records). The framework does not explain the numerical value; this requires the specific capacity budget of our universe.

Aspect	Standard QM	This Paper
$\hbar$ status	Postulate	Derived necessity (MD of PLEC)
ε value	$= \hbar$	Empirical (capacity budget)
Zero action	Excluded by $\hbar$	Excluded by structure (Thm 5.2)
Multiple records	$n\hbar$ (postulated)	$n\varepsilon$ (proved, Prop. 6.1)
History counting	Not addressed	$k!$ sectors from T_9
Lagrangian form	Written down, symmetry-constrained	Spectral action = partition function (Thm 10
Gauge group	Input	Tannaka–Krein recovery from superselection
Gravity	Input (Einstein–Hilbert)	Derived (Seeley–DeWitt a_2 coefficient)
Cosmological constant	Free parameter	Derived (a_0 coefficient; $3\pi/102^{61}$)

23. What Would Falsify This Result

F1 *Zero-action record formation.* Any record that forms at zero cost falsifies Theorem 5.2. Not observed.

F2 *Infinite capacity.* Any interface supporting unbounded distinctions falsifies A1. Not observed.

F3 *Classical mechanics at all scales.* Any regime where action is arbitrarily small falsifies the structural minimum. Not observed; quantum regime is universal at $S \sim \varepsilon$.

F4 *Spectral action $\neq$ partition function.* Direct computation shows they are identical (Theorem 10.1). A mismatch at any heat-kernel order would falsify the internalization. No such mismatch at coefficient orders a_0, a_2, a_4.

F5 *Geometric internalization failure.* Any of the chains $L_{\text{kolmogorov}} \to L_{\text{chartability}} \to L_{\text{lovelock}} \to L_{\text{coleman_mandula}}$ failing would force importation of external theorems, undermining the self-contained derivation. Verified numerically via the code checks.

F6 *APF outside its validity window.* Any phenomenon where $kT \sim \varepsilon^*$ holds and APF still applies, or where $kT \ll \varepsilon^*$ holds and APF fails, would falsify the regime-of-validity statement of §8. The Planck-era breakdown is consistent with the conventional expectation.

F7 *BSM extensions.* Specific observational outcomes — direct detection of a dark-matter particle outside the sterile-state window; CMB inflation parameters incompatible with staged relaxation; baryogenesis quantification inconsistent with the two-mechanism CP structure — would falsify specific BSM extensions. These are flagged as open, not claimed.

24. Summary

24.1 The central result

Zero accumulated action is structurally inadmissible for record-forming transitions (Theorem 5.2). The minimum ε is the MD component of PLEC made quantitative. The Standard Model Lagrangian with Einstein–Hilbert gravity is the Seeley–DeWitt heat-kernel expansion of the APF partition function (Theorem 10.1). Spacetime is forced (Part IV). One-loop flow, neutrino masses, spin-statistics, and the gauge group follow arithmetically from the APF-derived spectrum (Part V).

24.2 Physical implications

Implication	Explanation	Status
Minimum action exists	$\varepsilon > 0$ structurally forced (MD of PLEC)	**[P]**
Classical mechanics fails at small S	Assumes $S \to 0$ possible	**[P]**
QM is necessary near ε	Effective description at threshold	**[P_stru**
Measurements cost action	Record formation requires ε	**[P]**
n records cost $n\varepsilon$	Independent records sum	**[P]**
$k!$ history sectors	From T_9 append maps	**[P]**
APF partition function is (generalised) Fermi–Dirac	Binomial identity over admissible subsets	**[P]**
APF valid for $kT \ll \varepsilon^*$	Gapped ground state	**[P]**
Saturation entropy robust to corrections	Z_{sat} factorisation	**[P]**
Spectral action = APF partition function	$L_{\text{spectral_action_internal}}$	**[P]**
SM Lagrangian from Seeley–DeWitt	Heat-kernel expansion	**[P]**
Einstein equations unique in $d = 4$	A9.1–A9.5 + index counting	**[P]**
$G = \text{Poincaré} \times G_{\text{gauge}}$	Independent chains + L_{loc}	**[P]**
Full metric from causal order + A1	HKM + Malament internalized	**[P]**
One-loop beta coefficients	Casimir arithmetic	**[P]**
Type-I seesaw $m_\nu = -M_D M_R^{-1} M_D^T$	Block diagonalization	**[P]**
Gauge group SU(3) × SU(2) × U(1)	Tannaka–Krein from superselection	**[P]**
$\hbar$ is empirical value of ε	Numerical value from observation	Empirica
BSM mechanisms present	Dark matter, inflation, baryogenesis outlines	**[P_stru**

24.3 Conceptual status

Quantization of action: derived, not discreteness, not quantum-specific, structurally necessary. Lagrangian form: derived by spectral-action = partition-function identity, not written down. Geometric structure: internalized, with no external theorem dependencies in the core chain. The framework has internalized, in Paper 7 v2.0, the last external inputs of the derivation: all 11 external theorems used in prior spectral-action treatments are now APF-internal. The remaining empirical inputs are the scale of ε (Planck's constant), the value of σ_0 (Higgs VEV), and the top/bottom quark absolute-mass scales (Type B anchors in T_{m}).

24.4 Paper 3 v3.0 bridge

The saturation factorisation of §9 provides the thermodynamic formalism for Paper 3 v3.0's ledger-accounting principle. The forward reference in Paper 3 v3.0 §5 is answered here: entropy-as-count is robust under cost-landscape corrections because Z_{sat} factorises into a pure count N_{micro} and an additive energy shift $E_{\text{sat}}(\eta)$ that absorbs thermal and interaction corrections.

A. Framework Cost Parameters: ε, κ, η

Parameter	Value	Role in This Paper	Status
ε (T_ε)	> 0	Minimum enforcement cost = abstract ε = MD	**[P]**
κ (T_κ)	$= 2$	Per-channel cost = 2ε (forward + backward)	**[P]**
η (T_η)	$\leq \varepsilon$	Cross-correlation bound; enters $V(\Phi)$ and $E_{\text{sat}}(\eta)$	**[P_structural]**
ε^*	$\sim E_P$	Capacity-type enforcement scale; sets $Z_0(\beta)$	**[P]**
d_{eff}	$= 102$	Internal states per capacity mode; $L_{\text{self_exclusion}}$	**[P]**
C_{total}	$= 61$	Total capacity types; $42 + 19$ partition from L_{count}	**[P]**

The abstract minimum ε of this paper is exactly the T_ε enforcement granularity of Paper 3. The per-channel cost $\kappa\varepsilon = 2\varepsilon$ (T_κ) means each distinguishable channel costs at least 2ε of action. The subordination bound $\eta \leq \varepsilon$ (T_η) enters the enforcement potential $V(\Phi)$ and absorbs the saturation-energy shift $E_{\text{sat}}(\eta)$ additively.

B. Toy-Model Verification of Minimum Action

Interfaces: $I = \{0, 1\}$. Capacities: $C_0 = C_1 = 10$. Distinctions: $D = \{d_1, d_2, d_3\}$. Demands: $\delta_0(d_1) = 2$, $\delta_1(d_1) = 1$, etc. Compute $\varepsilon = \min\{\|\delta(d)\|_1 : d \text{ is feasibly addable relative to some admissible } S\}$. For any admissible S and addable d: $\|\delta(d)\|_1 \geq 3$ (in this example). Therefore $\varepsilon = 3$. Any path adding a new distinction accumulates cost ≥ 3. Verified by enumeration over all admissible paths.

The $T0.4'$ BFS certificate (Paper 2) provides a more sophisticated verification on the canonical witness world, confirming that record removal is inadmissible by exhaustive graph search.

C. PLEC Assumption Inventory for Paper 7

For each theorem or lemma of Paper 7, the table indicates which PLEC components are used.

Theorem / Lemma	**A1**	**MD**	**A2**	**BW**
Lemma 5.1 (cost of introducing distinction)	•	•		
Theorem 5.2 (Isolation of Zero)	•	•		•
Proposition 6.1 (n-record bound)	•	•		
$k!$ history sectors (§6)	•	•		
$Z_0(\beta)$ derivation (§7)	•			
$kT \ll \varepsilon^*$ regime (§8)	•	•		
Z_{sat} factorisation (§9)	•			
Theorem 10.1 ($L_{\text{spectral_action_internal}}$)	•		•	•
$L_{\text{normalization_coefficient}}$			•	
$L_{\text{scalar_potential_form}}$	•		•	
$V(\Phi)$ and mass gap (§12)	•	•	•	
$L_{\text{kolmogorov_internal}}$	•			
$L_{\text{chartability}}$	•			
$L_{\text{lovelock_internal}}$	•		•	
$L_{\text{coleman_mandula_internal}}$	•			•
$L_{\text{HKM_causal_geometry}}$	•			
$L_{\text{Malament_uniqueness}}$	•			
$T_{6B,\text{beta_one_loop}}$	•			
$L_{\text{seesaw_type_I}}$	•			
$L_{\text{Pauli_Jordan}}$	•			
$L_{\text{McKean_Singer_internal}}$	•			
$L_{\text{Tannaka_Krein}}$	•		•	

All four PLEC components appear in the paper. A1 and MD dominate the foundational layer (Part I). A2 appears in the variational-selection arguments of the internalization layer (Part III). BW prevents degeneracy in the record-forming paths (Theorem 5.2) and in the direct-product structure (Part IV §16).

D. Changelog

Change	**Section**	**Details**
First .tex release	All	Feb-2026 PDF content converted to .tex (Part I).
PLEC framing	Abstract, §1	PLEC 4-component overlay; T_ε retagged as MD.
Partition function	Part II	New: $Z_0(\beta)$ derivation; $kT \ll \varepsilon^*$ regime; Z_{sat}.
Spectral action identity	§10	$L_{\text{spectral_action_internal}}$: SM Lagrangian from heat kernel.
Scalar-potential internalization	§11	$L_{\text{normalization_coefficient}}$; $L_{\text{scalar_potential_form}}$.
Geometric internalization	Part IV	Kolmogorov, chartability, Lovelock, Coleman–Mandula, HKM
SM Lagrangian extensions	Part V	Beta one-loop, seesaw, Pauli–Jordan, McKean–Singer, Tannak
BSM mechanisms	§21	Dark matter, inflation, baryogenesis: mechanisms only, conser
Falsification expanded	§23	F4–F7 added (spectral identity, internalization, regime, BSM)
Planck's constant table	§22	Lagrangian / gauge / gravity / Λ rows added.
Implications table	§24	14 new rows.
PLEC assumption inventory	App. C	New appendix: A1/MD/A2/BW usage per theorem.
Date	Header	Feb 2026 → April 2026.
Accompanying doc	—	Technical Supplement v1.0 released concurrently (full formal p

Paper 7 — Action (v2.0) — Admissibility Physics — April 2026

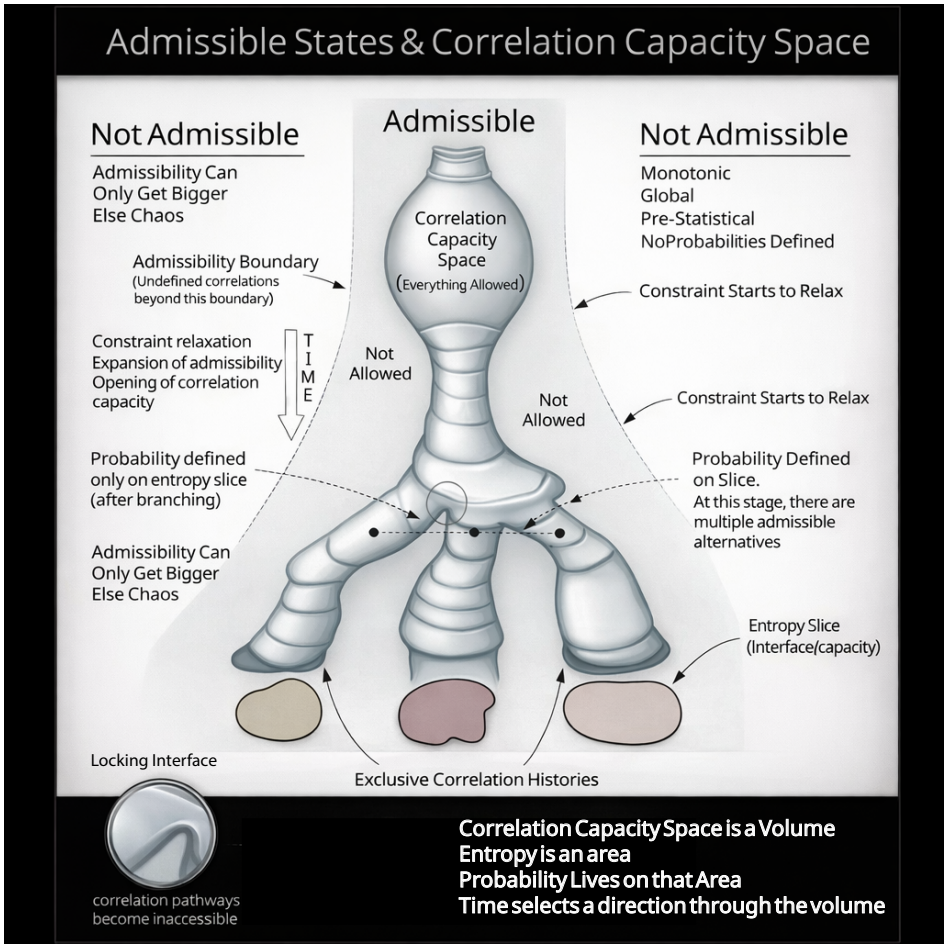

Figure 1: **What an admissible path actually is.** The upper bulb is the admissible class $\mathcal{A}_\Gamma$ (the bounded region in correlation-capacity space carved out by A1 and MD). An admissible path $P = \{S_\lambda\}$ is a continuous trajectory through this volume: at each step, the configuration S_λ stays inside the boundary (failure to do so is the case the flanking "Not Admissible" regions warn against). The downward arrow labeled "time" is exactly the direction along which the accumulated cost $A(P) = \sum_i \mathrm{Var}_\lambda[\cdots]$ of §3 is measured: the action functional in this paper is the integral of enforcement-load change along the trajectory shown. The narrowing near the bottom is where the Minimum Action Theorem (§5) becomes load-bearing: any record-forming path must commit at least ε of accumulated cost, ruling out infinitesimal actions and isolating zero from the action spectrum. The locking interface and three downstream tubes visualize the $k!$ history sectors of §6 (formalized in *Code reference:* `check_T9` in `gauge.py`.); the irreversibility of the lock is the L_{irr} mechanism of *Code reference:* `check_L_irr` in `core.py`. — locally unrecoverable correlation, not infinite reverse cost. The "volume / area" note at the bottom is the Bekenstein-area structure that the Technical Supplement uses to identify the Connes spectral action with the APF partition function: entropy on a slice scales with interface area, not bulk volume.